



Amazon

MARKETPLACE ANALYTICS Brazil



CUSTOMER INSIGHTS, REVENUE INTELLIGENCE &
OPERATIONAL PERFORMANCE USING SQL



TOOLS USED IN SQL :

- *JOINS
- *SUBQUERIES
- *WINDOW
FUNCTIONS
- *BUSINESS
ANALYTICS





Project Overview

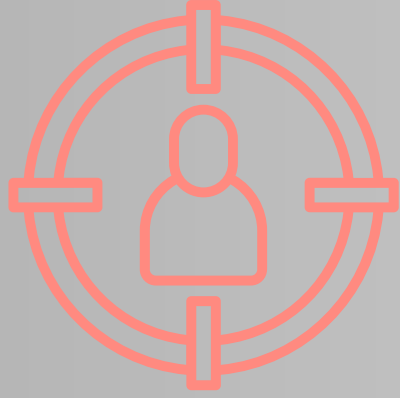


Amazon Brazil is one of Latin America's fastest-growing e-commerce platforms. As the marketplace continues to scale, understanding customer behavior, seller performance, revenue generation, and operational efficiency becomes increasingly important. This project leverages SQL to analyze transactional data across customers, orders, products, sellers, payments, and reviews to generate actionable business insights.



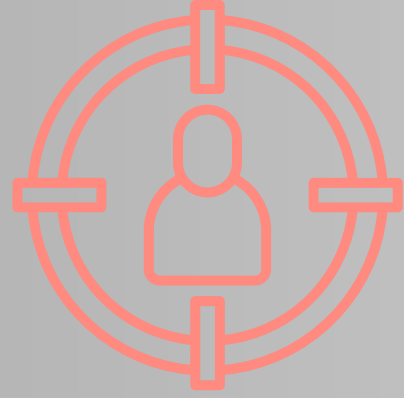
PROJECT OBJECTIVES

- • ANALYZE CUSTOMER PURCHASING BEHAVIOR
- • EVALUATE SELLER PERFORMANCE
- • IDENTIFY HIGH-PERFORMING PRODUCT CATEGORIES
- • ASSESS PAYMENT TRENDS
- • MEASURE DELIVERY PERFORMANCE
- • IMPROVE CUSTOMER SATISFACTION



Problem Statements:

- **Q1. FIND THE TOTAL NUMBER OF ORDERS FULFILLED BY EACH SELLER STATE.**
- **Q2. FOR EACH PRODUCT CATEGORY, CALCULATE THE CUMULATIVE REVENUE GENERATED AS ORDERS COME IN OVER TIME.**
- **Q3. WHICH PAYMENT METHOD DO CUSTOMERS USE THE MOST, AND WHAT IS THE AVERAGE ORDER VALUE FOR EACH PAYMENT TYPE?**
- **Q4. FIND THE CUSTOMER WHO HAS SPENT THE MOST MONEY ACROSS ALL THEIR ORDERS.**
- **Q5. FIND THE AVERAGE REVIEW SCORE FOR EACH PRODUCT CATEGORY.**
- **Q6. FIND THE TOTAL NUMBER OF ORDERS PLACED BY EACH CUSTOMER, BROKEN DOWN BY THE STATE THEY LIVE IN.**
- **Q7. IDENTIFY SELLERS WHO REGISTERED ON THE PLATFORM BUT HAVE NEVER FULFILLED A SINGLE ORDER.**



Problem Statements:

- **Q8. FIND THE TOP 5 PRODUCT CATEGORIES BY TOTAL REVENUE.**
- **Q9. FIND THE MEDIAN DELIVERY TIME (IN DAYS) BETWEEN ORDER PLACEMENT AND ACTUAL DELIVERY.**
- **Q10. FIND ALL PRODUCTS THAT HAVE NEVER BEEN ORDERED.**
- **Q11. FIND SELLERS WHO HAVE FULFILLED MORE ORDERS THAN THE AVERAGE SELLER ON THE PLATFORM.**
- **Q12. FIND WHICH BRAZILIAN STATES HAVE THE HIGHEST AVERAGE CUSTOMER REVIEW SCORE FOR ORDERS DELIVERED THERE.**
- **Q13. IDENTIFY CUSTOMERS WHO HAVE PLACED ORDERS BUT NEVER LEFT A REVIEW.**
- **Q14. FIND THE MONTH WITH THE HIGHEST NUMBER OF ORDERS PLACED ACROSS THE ENTIRE PLATFORM.**

1. Total Number of Orders Fulfilled by Each Seller State

BUSINESS CONTEXT	• FULFILLMENT CENTER INVESTMENT • SELLER STATE ANALYSIS • ORDER VOLUME DISTRIBUTION • LOGISTICS INFRASTRUCTURE PLANNING	BUSINESS OBJECTIVE	• MEASURE ORDERS BY STATE • IDENTIFY HIGH-DEMAND REGIONS • EVALUATE SELLER PERFORMANCE • SUPPORT EXPANSION DECISIONS
SQL QUERY AND OUTPUT	<pre>SELECT s.seller_state, COUNT(DISTINCT oi.order_id) AS total_orders FROM sellers s INNER JOIN order_items oi ON s.seller_id = oi.seller_id GROUP BY s.seller_state ORDER BY total_orders DESC;</pre> SELLERS (SELLER_ID → SELLER_STATE) → INNER JOIN ORDER_ITEMS (MATCHES ON SELLER_ID) → COUNT DISTINCT UNIQUE ORDER_ID VALUES → GROUP BY STATE → ORDER BY VOLUME DESCENDING.		
KEY INSIGHTS	• SELLER CONCENTRATION • REGIONAL DEMAND DISTRIBUTION • HIGH-VOLUME SELLER HUBS • MARKETPLACE ACTIVITY TRENDS	BUSINESS IMPACT	• DATA-DRIVEN EXPANSION • WAREHOUSE LOCATION PLANNING • IMPROVED DELIVERY EFFICIENCY • REGIONAL RESOURCE ALLOCATION

2. Cumulative Revenue Generated by Product Category

BUSINESS CONTEXT	• REVENUE GROWTH TRACKING • CATEGORY PERFORMANCE ANALYSIS • REVENUE TREND MONITORING • PROMOTIONAL CAMPAIGN PLANNING	BUSINESS OBJECTIVE	• CALCULATE RUNNING REVENUE • COMPARE CATEGORY GROWTH • IDENTIFY FAST-GROWING CATEGORIES • SUPPORT MARKETING DECISIONS
SQL QUERY AND OUTPUT	<pre>SELECT p.product_category_name, o.order_purchase_timestamp, oi.order_id, SUM(oi.price) OVER (PARTITION BY p.product_category_name ORDER BY o.order_purchase_timestamp ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS cumulative_revenue FROM products p JOIN order_items oi ON p.product_id = oi.product_id JOIN orders o ON oi.order_id = o.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') AND p.product_category_name IS NOT NULL -- ◀ ADD THIS LINE ORDER BY p.product_category_name, o.order_purchase_timestamp;</pre>		
KEY INSIGHTS	• CATEGORY REVENUE GROWTH • REVENUE ACCUMULATION TRENDS • FAST-GROWING CATEGORIES	BUSINESS IMPACT	• BETTER PROMOTION PLANNING • TARGETED FLASH SALES • INVENTORY OPTIMIZATION • REVENUE FORECASTING

3. Most Used Payment Method & Average Order Value by Payment Type

BUSINESS CONTEXT	• PAYMENT METHOD ANALYSIS • CUSTOMER PAYMENT PREFERENCES • FINANCIAL PERFORMANCE TRACKING • PAYMENT PROVIDER EVALUATION	BUSINESS OBJECTIVE	• IDENTIFY DOMINANT PAYMENT METHOD • CALCULATE AVERAGE ORDER VALUE • ANALYZE CUSTOMER SPENDING PATTERNS • SUPPORT PROVIDER NEGOTIATIONS
SQL QUERY AND OUTPUT	<pre>SELECT p.payment_type, COUNT(DISTINCT p.order_id) AS total_orders, ROUND(AVG(p.payment_value), 2) AS average_order_value FROM order_payments p JOIN orders o ON p.order_id = o.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') GROUP BY p.payment_type ORDER BY total_orders DESC;</pre> ORDER_PAYMENTS (PAYMENT_TYPE) → JOIN ORDERS (FILTER ACTIVE STATUS) → COUNT DISTINCT UNIQUE ORDER_ID VALUES → AVG PAYMENT_VALUE PER TYPE → GROUP BY METHOD → ORDER BY VOLUME DESCENDING.		
KEY INSIGHTS	• MOST PREFERRED PAYMENT METHOD • CUSTOMER SPENDING BEHAVIOR	BUSINESS IMPACT	• PAYMENT PARTNER NEGOTIATIONS • CHECKOUT EXPERIENCE OPTIMIZATION • REVENUE GROWTH

4. Highest Spending Customer on orders.

BUSINESS CONTEXT	• CUSTOMER LOYALTY PROGRAM • VIP CUSTOMER IDENTIFICATION • CUSTOMER VALUE ANALYSIS • PREMIUM TIER ONBOARDING	BUSINESS OBJECTIVE	• IDENTIFY TOP CUSTOMER • CALCULATE TOTAL SPENDING • ANALYZE CUSTOMER VALUE • SUPPORT RETENTION STRATEGY
SQL QUERY AND OUTPUT	<pre>SELECT c.customer_unique_id, COUNT(DISTINCT o.order_id) AS total_orders_placed, ROUND(SUM(p.payment_value), 2) AS total_lifetime_spend FROM customers c JOIN orders o ON c.customer_id = o.customer_id JOIN order_payments p ON o.order_id = p.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') GROUP BY c.customer_unique_id ORDER BY total_lifetime_spend DESC LIMIT 1;</pre> CUSTOMERS (CUSTOMER_UNIQUE_ID) → JOIN ORDERS (VIA CUSTOMER_ID) → JOIN ORDER_PAYMENTS (VIA ORDER_ID) → SUM PAYMENT_VALUE → GROUP BY UNIQUE ID → ORDER BY SPEND DESCENDING → LIMIT 1.		
KEY INSIGHTS	• HIGHEST REVENUE CUSTOMER • PREMIUM CUSTOMER SEGMENT	BUSINESS IMPACT	• VIP LOYALTY PROGRAM DESIGN • CUSTOMER RETENTION STRATEGY • PERSONALIZED ENGAGEMENT • REVENUE MAXIMIZATION

5. Average Review Score by Product Category

BUSINESS CONTEXT	• CUSTOMER SATISFACTION ANALYSIS • PRODUCT QUALITY MONITORING • CATEGORY PERFORMANCE EVALUATION • CUSTOMER EXPERIENCE IMPROVEMENT	BUSINESS OBJECTIVE	• MEASURE CATEGORY SATISFACTION • CALCULATE AVERAGE REVIEW SCORES • IDENTIFY LOW-RATED CATEGORIES • PRIORITIZE IMPROVEMENT AREAS
SQL QUERY AND OUTPUT	<pre>SELECT p.product_category_name, ROUND(AVG(r.review_score), 2) AS average_review_score, COUNT(DISTINCT o.order_id) AS total_orders_reviewed FROM products p JOIN order_items oi ON p.product_id = oi.product_id JOIN orders o ON oi.order_id = o.order_id JOIN order_reviews r ON o.order_id = r.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') AND p.product_category_name IS NOT NULL GROUP BY p.product_category_name ORDER BY average_review_score ASC, total_orders_reviewed DESC;</pre> PRODUCTS (PRODUCT_CATEGORY_NAME) → JOIN ORDER_ITEMS → JOIN ORDERS → JOIN ORDER_REVIEWS (REVIEW_SCORE) → AVG SCORE PER GROUP → FILTER OUT NULLS → ORDER BY SCORE ASCENDING.		
KEY INSIGHTS	• CUSTOMER SATISFACTION LEVELS • BEST-RATED CATEGORIES • LOW-PERFORMING	BUSINESS IMPACT	• CATEGORY IMPROVEMENT PLANNING • PRODUCT QUALITY ENHANCEMENT • BETTER CUSTOMER EXPERIENCE

6. Total Orders Placed by Each Customer by State

BUSINESS CONTEXT	• REGIONAL MARKETING ANALYSIS • CUSTOMER PURCHASE BEHAVIOR • STATE-WISE DEMAND TRACKING • CUSTOMER ENGAGEMENT STRATEGY	BUSINESS OBJECTIVE	• MEASURE ORDER FREQUENCY • ANALYZE STATE-WISE ACTIVITY • IDENTIFY HIGH-VALUE REGIONS • SUPPORT MARKETING CAMPAIGNS
SQL QUERY AND OUTPUT	<pre>SELECT c.customer_state, c.customer_unique_id, COUNT(DISTINCT o.order_id) AS total_orders_placed FROM customers c JOIN orders o ON c.customer_id = o.customer_id WHERE o.order_status NOT IN ('canceled', 'unavailable') GROUP BY c.customer_state, c.customer_unique_id ORDER BY c.customer_state ASC, total_orders_placed DESC;</pre> CUSTOMERS (CUSTOMER_STATE + CUSTOMER_UNIQUE_ID) → JOIN ORDERS (MATCHES ON CUSTOMER_ID) → FILTER OUT INACTIVE SALES → COUNT DISTINCT UNIQUE ORDER_ID VALUES → GROUP BY STATE THEN UNIQUE ID → ORDER BY STATE AND HIGHEST FREQUENCY.		
KEY INSIGHTS	• STATE-WISE ORDER PATTERNS • CUSTOMER PURCHASE FREQUENCY • HIGH-ACTIVITY REGIONS • CUSTOMER ENGAGEMENT	BUSINESS IMPACT	• TARGETED MARKETING CAMPAIGNS • LOYALTY PROGRAM OPTIMIZATION • REGIONAL GROWTH STRATEGY • CUSTOMER RETENTION PLANNING

7. Sellers Who Have registered but Never Fulfilled a Single Order

BUSINESS CONTEXT	• DORMANT SELLER IDENTIFICATION • SELLER ENGAGEMENT STRATEGY • MARKETPLACE GROWTH MANAGEMENT • SELLER ACCOUNT OPTIMIZATION	BUSINESS OBJECTIVE	• IDENTIFY INACTIVE SELLERS • DETECT ZERO-ORDER ACCOUNTS • SUPPORT RE-ENGAGEMENT CAMPAIGNS • IMPROVE SELLER NETWORK EFFICIENCY
SQL QUERY AND OUTPUT	<pre>SELECT s.seller_id, s.seller_city, s.seller_state FROM sellers s LEFT JOIN order_items oi ON s.seller_id = oi.seller_id LEFT JOIN orders o ON oi.order_id = o.order_id AND o.order_status NOT IN ('canceled', 'unavailable') WHERE oi.seller_id IS NULL ORDER BY s.seller_state ASC, s.seller_city ASC;</pre> SELLERS (SELLER_ID) → LEFT JOIN ORDER_ITEMS → WHERE LINE ITEM LINK IS NULL → GROUP / ORDER BY LOCATION TO BUILD THE DORMANT ACCOUNT LIST.		
KEY INSIGHTS	• DORMANT SELLER ACCOUNTS • ZERO FULFILLMENT ACTIVITY • SELLER PARTICIPATION GAPS • MARKETPLACE UTILIZATION ISSUES	BUSINESS IMPACT	• TARGETED RE-ENGAGEMENT CAMPAIGNS • SELLER NETWORK OPTIMIZATION • ACCOUNT REVIEW & CLEANUP • IMPROVED MARKETPLACE

8. Find the top 5 product categories by total revenue.

BUSINESS CONTEXT • REVENUE PERFORMANCE ANALYSIS • CATEGORY PRIORITIZATION • PROMOTIONAL BUDGET ALLOCATION • SALES EVENT PLANNING	BUSINESS OBJECTIVE • CALCULATE CATEGORY REVENUE • IDENTIFY TOP REVENUE DRIVERS • RANK PRODUCT CATEGORIES • SUPPORT MERCHANDISING DECISIONS
SQL QUERY AND OUTPUT <pre>SELECT p.product_category_name, ROUND(SUM(oi.price), 2) AS total_revenue, COUNT(DISTINCT oi.order_id) AS total_orders_placed FROM products p JOIN order_items oi ON p.product_id = oi.product_id JOIN orders o ON oi.order_id = o.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') AND p.product_category_name IS NOT NULL GROUP BY p.product_category_name ORDER BY total_revenue DESC LIMIT 5;</pre>	PRODUCTS (PRODUCT_CATEGORY_NAME) → JOIN ORDER_ITEMS (PRICE) → JOIN ORDERS (FILTER ACTIVE STATUS) → SUM PRICE PER GROUP → FILTER OUT NULLS → ORDER BY REVENUE DESCENDING → LIMIT 5.
KEY INSIGHTS • HIGHEST REVENUE CATEGORIES • MAJOR SALES CONTRIBUTORS • REVENUE CONCENTRATION TRENDS	BUSINESS IMPACT • PROMOTIONAL BUDGET OPTIMIZATION • HOMEPAGE PLACEMENT STRATEGY • INVENTORY PLANNING • SALES EVENT PREPARATION

Q9. Median Delivery Time (in days) Between Order Placement and Actual Delivery

**BUSINESS
CONTEXT**

- DELIVERY PERFORMANCE ANALYSIS
- CUSTOMER EXPERIENCE MONITORING
- LOGISTICS EFFICIENCY EVALUATION
- SERVICE LEVEL OPTIMIZATION

**BUSINESS
OBJECTIVE**

- MEASURE MEDIAN DELIVERY TIME
- ASSESS DELIVERY PERFORMANCE
- REDUCE CUSTOMER WAIT TIME
- SUPPORT LOGISTICS IMPROVEMENTS

**SQL
QUERY
AND
OUTPUT**

```
WITH delivery_time AS (  
    SELECT  
        DATEDIFF(  
            o.order_delivered_customer_date,  
            o.order_purchase_timestamp  
        ) AS delivery_days  
    FROM orders o  
    WHERE o.order_delivered_customer_date IS NOT NULL  
)  
SELECT AVG(delivery_days) AS median_delivery_days
```

```
FROM (  
    SELECT  
        delivery_days,  
        ROW_NUMBER() OVER (ORDER BY delivery_days) AS rn  
        COUNT(*) OVER () AS total_rows  
    FROM delivery_time  
) t  
WHERE rn IN (  
    FLOOR((total_rows + 1)/2),  
    FLOOR((total_rows + 2)/2)  
);
```

ORDERS
(DATEDIFF) →
CTE COMPUTES
SHIPPING DAYS
→ SUBQUERY
INDEXES AND
COUNTS ROWS →
FILTER TARGETS
MIDDLE INDEX →
AVG
CALCULATES
TRUE MEDIAN.
RUN THE SCRIPT
TO SEE IF YOUR
RESULT HITS 10
DAYS!

**KEY
INSIGHTS**

- TYPICAL CUSTOMER WAIT TIME
- DELIVERY CONSISTENCY LEVELS
- LOGISTICS PERFORMANCE TRENDS

**BUSINESS
IMPACT**

- LOGISTICS NETWORK OPTIMIZATION
- IMPROVED CUSTOMER SATISFACTION
- FASTER DELIVERY PLANNING

10. Find all products that have never been ordered.

BUSINESS CONTEXT	• CATALOG CLEANUP • PRODUCT PERFORMANCE MONITORING • INVENTORY OPTIMIZATION • LISTING QUALITY MANAGEMENT	BUSINESS OBJECTIVE	• IDENTIFY UNORDERED PRODUCTS • DETECT DEAD LISTINGS • IMPROVE CATALOG EFFICIENCY • SUPPORT LISTING OPTIMIZATION
SQL QUERY AND OUTPUT	<pre>SELECT p.product_id, p.product_category_name FROM products p LEFT JOIN order_items oi ON p.product_id = oi.product_id WHERE oi.product_id IS NULL;</pre> PRODUCTS (PRODUCT_ID) → LEFT JOIN ORDER_ITEMS → WHERE LINE ITEM LINK IS NULL → SELECTS UNPURCHASED CATALOG ITEMS.		
KEY INSIGHTS	• UNORDERED PRODUCTS • DEAD CATALOG LISTINGS • LOW PRODUCT VISIBILITY • CATEGORY OPTIMIZATION OPPORTUNITIES	BUSINESS IMPACT	• CATALOG CLEANUP • SEARCH EXPERIENCE IMPROVEMENT • INVENTORY OPTIMIZATION • BETTER PRODUCT

11. Sellers Who Have Fulfilled More Orders Than the Average Seller			
BUSINESS CONTEXT	• SELLER PERFORMANCE ANALYSIS • TOP SELLER IDENTIFICATION • BADGE PROGRAM QUALIFICATION • SELLER SUCCESS EVALUATION	BUSINESS OBJECTIVE	• MEASURE SELLER PERFORMANCE • CALCULATE AVERAGE ORDERS • IDENTIFY ABOVE-AVERAGE SELLERS • SUPPORT BADGE PROGRAM SELECTION
SQL QUERY AND OUTPUT	<pre>WITH seller_orders AS (SELECT oi.seller_id, COUNT(DISTINCT oi.order_id) AS total_orders FROM order_items oi JOIN orders o ON oi.order_id = o.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') GROUP BY oi.seller_id) SELECT seller_id, total_orders FROM seller_orders WHERE total_orders > (SELECT AVG(total_orders) FROM seller_orders) ORDER BY total_orders DESC;</pre> ORDER_ITEMS (COUNT DISTINCT ORDER_ID) → CTE AGGREGATES TOTAL ORDERS PER SELLER_ID → SUBQUERY CALCULATES GLOBAL AVERAGE ORDERS → WHERE FILTERS SELLERS CLEARING THAT AVERAGE BENCHMARK → ORDER BY HIGHEST VOLUME.		
KEY INSIGHTS	• HIGH-PERFORMING SELLERS • ABOVE-AVERAGE ORDER VOLUME • SELLER PERFORMANCE DISTRIBUTION • TOP SELLER CANDIDATES	BUSINESS IMPACT	• TOP SELLER BADGE SELECTION • SELLER RECOGNITION PROGRAMS • PERFORMANCE BENCHMARKING • MARKETPLACE GROWTH SUPPORT

12. Brazilian States with the Highest Average Customer Review Score for Delivered Orders			
BUSINESS CONTEXT	• BRAZILIAN STATE PERFORMANCE • DELIVERY EXPERIENCE ANALYSIS • REGIONAL CUSTOMER SATISFACTION • LAST-MILE LOGISTICS EVALUATION	BUSINESS OBJECTIVE	• MEASURE REVIEW SCORES BY STATE • IDENTIFY TOP-RATED BRAZILIAN STATES • EVALUATE DELIVERY SATISFACTION • IMPROVE REGIONAL LOGISTICS PERFORMANCE
SQL QUERY AND OUTPUT	<pre>SELECT c.customer_state, ROUND(AVG(r.review_score), 2) AS average_review_score, COUNT(DISTINCT o.order_id) AS total_orders_delivered FROM customers c JOIN orders o ON c.customer_id = o.customer_id JOIN order_reviews r ON o.order_id = r.order_id WHERE o.order_status = 'delivered' GROUP BY c.customer_state ORDER BY average_review_score DESC;</pre> CUSTOMERS (CUSTOMER_STATE) → JOIN ORDERS (FILTER DELIVERED) → JOIN ORDER_REVIEWS (REVIEW_SCORE) → AVG SCORE PER STATE → ORDER BY SCORE DESCENDING.		
KEY INSIGHTS	• BRAZILIAN STATE SATISFACTION RANKING • REGIONAL SERVICE QUALITY TRENDS • HIGH-PERFORMING DELIVERY	BUSINESS IMPACT	• LOGISTICS PARTNER EVALUATION • REGIONAL SERVICE IMPROVEMENTS • CUSTOMER SATISFACTION ENHANCEMENT

13. Identify customers who have placed orders but never left a review.

BUSINESS CONTEXT	• CUSTOMER ENGAGEMENT ANALYSIS • REVIEW PARTICIPATION TRACKING • POST-PURCHASE EXPERIENCE • CUSTOMER FEEDBACK GROWTH	BUSINESS OBJECTIVE	• IDENTIFY NON-REVIEWING CUSTOMERS • INCREASE REVIEW VOLUME • IMPROVE CUSTOMER ENGAGEMENT • SUPPORT FEEDBACK COLLECTION
SQL QUERY AND OUTPUT	<pre>SELECT c.customer_unique_id, c.customer_id, o.order_id FROM customers c JOIN orders o ON c.customer_id = o.customer_id LEFT JOIN order_reviews r ON o.order_id = r.order_id WHERE o.order_status NOT IN ('canceled', 'unavailable') AND r.order_id IS NULL;</pre> CUSTOMERS → JOIN ORDERS (ACTIVE RECORDS) → LEFT JOIN ORDER_REVIEWS → WHERE REVIEW LINK IS NULL → EXTRACTS NON-REVIEWING CUSTOMERS.		
KEY INSIGHTS	• LOW REVIEW PARTICIPATION • CUSTOMER ENGAGEMENT GAPS • UNTAPPED FEEDBACK OPPORTUNITIES • POST-PURCHASE BEHAVIOR	BUSINESS IMPACT	• REVIEW REMINDER CAMPAIGNS • INCREASED REVIEW GENERATION • ENHANCED PRODUCT DISCOVERABILITY • IMPROVED CUSTOMER TRUST

14. Find the month with the highest number of orders placed across the entire platform.

BUSINESS CONTEXT	• DEMAND FORECASTING • PEAK SEASON ANALYSIS • CAPACITY PLANNING • WORKFORCE MANAGEMENT	BUSINESS OBJECTIVE	• IDENTIFY PEAK ORDER MONTH • MEASURE MONTHLY DEMAND • SUPPORT RESOURCE PLANNING • OPTIMIZE SEASONAL OPERATIONS
SQL QUERY AND OUTPUT	<pre>SELECT DATE_FORMAT(order_purchase_timestamp, '%Y-%m') AS order_month, COUNT(DISTINCT order_id) AS total_orders FROM orders WHERE order_status NOT IN ('canceled', 'unavailable') GROUP BY order_month ORDER BY total_orders DESC LIMIT 1;</pre>		ORDERS (FILTER ACTIVE) → DATE_FORMAT EXTRACTS YEAR-MONTH → COUNT DISTINCT UNIQUE ORDERS → GROUP BY MONTH → ORDER BY VOLUME DESCENDING → LIMIT 1
KEY INSIGHTS	• PEAK DEMAND PERIOD • MONTHLY ORDER TRENDS • SEASONAL PURCHASING PATTERNS • HIGH-TRAFFIC SALES WINDOWS	BUSINESS IMPACT	• WAREHOUSE CAPACITY PLANNING • WORKFORCE ALLOCATION • MARKETING CAMPAIGN SCHEDULING • DEMAND FORECASTING



KEY ACHIEVEMENTS

BUSINESS PERFORMANCE ANALYSIS

- ✓ IDENTIFIED TOP REVENUE-GENERATING PRODUCT CATEGORIES
- ✓ TRACKED CUMULATIVE REVENUE GROWTH ACROSS CATEGORIES

CUSTOMER ANALYTICS

- ✓ IDENTIFIED HIGHEST-VALUE CUSTOMER SEGMENTS
- ✓ ANALYZED CUSTOMER ORDERING BEHAVIOR BY REGION
- ✓ IDENTIFIED CUSTOMERS WITH LOW REVIEW PARTICIPATION

MARKETPLACE OPTIMIZATION

- ✓ IDENTIFIED UNUSED PRODUCTS WITH ZERO ORDER HISTORY
- ✓ ANALYZED PAYMENT PREFERENCES AND TRANSACTION BEHAVIOR

CUSTOMER EXPERIENCE & OPERATIONS

- ✓ MEASURED CUSTOMER SATISFACTION ACROSS PRODUCT CATEGORIES
- ✓ ASSESSED REVIEW PERFORMANCE ACROSS BRAZILIAN STATES





STRATEGIC BUSINESS ACTIONS

REVENUE OPTIMIZATION

- PRIORITIZE INVESTMENT IN TOP-PERFORMING PRODUCT CATEGORIES
- ALIGN PROMOTIONAL CAMPAIGNS WITH CATEGORY REVENUE TRENDS

CUSTOMER GROWTH & RETENTION

- DEVELOP PREMIUM LOYALTY PROGRAMS FOR HIGH-VALUE CUSTOMERS
- INCREASE REVIEW PARTICIPATION THROUGH TARGETED ENGAGEMENT INITIATIVES

LOGISTICS & SERVICE EXCELLENCE

- EXPAND FULFILLMENT INFRASTRUCTURE IN HIGH-DEMAND REGIONS
- IMPROVE DELIVERY PERFORMANCE IN LOWER-RATED BRAZILIAN STATES

MARKETPLACE OPTIMIZATION

- REVIEW, OPTIMIZE, OR REMOVE PRODUCTS WITH NO SALES HISTORY
- PREPARE INVENTORY AND WORKFORCE CAPACITY FOR PEAK-DEMAND PERIODS



Thankyou

